Subject: Mathematics | **PERIODIC TEST – 05 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks:** 30

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

A two-digit number with distinct digits is formed using the digits 2, 3, and 5. The probability that the number formed is divisible by 5 is:

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{1}{6}$

Working & Calculation:

- Digits available: $\{2, 3, 5\}$.
- Two-digit numbers formed with distinct digits: Ten's place has 3 choices; unit's place has 2 remaining choices $\implies n(S) = 3 \times 2 = 6$.

$$S = \{23, 25, 32, 35, 52, 53\}$$

- A number is divisible by 5 if its units digit is 5:

$$E = \{25, 35\} \implies n(E) = 2$$

- Probability:

$$P(\text{divisible by } 5) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{3}$

Question 2 [1 Mark]

In a lottery, there are 10 prizes and 25 blanks. The probability of getting a prize is:

(a) $\frac{2}{7}$ (b) $\frac{5}{7}$
(c) $\frac{2}{5}$ (d) $\frac{1}{3}$

Working & Calculation:

- Total tickets in the lottery: $n(S) = 10$ (prizes) $+ 25$ (blanks) $= 35$ tickets.
- Number of prize-bearing tickets: $n(E) = 10$.
- Probability of getting a prize:

$$P(\text{prize}) = \frac{n(E)}{n(S)} = \frac{10}{35} = \frac{2}{7}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{2}{7}$

Question 3 [1 Mark]

If the probability of winning a game is 0.07, the probability of losing it is:

(a) **0.93** (b) **0.03**

(c) **0.70** (d) **0.97**

Working & Calculation:

- Winning the game (E) and losing the game ($\bar{E}$) are complementary events:

$$P(E) + P(\bar{E}) = 1$$

- Given $P(\text{winning}) = 0.07$:

$$P(\text{losing}) = 1 - P(\text{winning}) = 1 - 0.07 = 0.93$$

CORRECT ANSWER: OPTION (a) $\longrightarrow$ 0.93

Question 4 [1 Mark]

A target is in the shape of a circle of radius r enclosed inside a square of side $2r$. A dart is thrown at random and lands inside the square. The probability that it lands inside the circle is:

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) $\frac{1}{\pi}$ (d) $\frac{\pi}{8}$

Working & Calculation:

- Total area of the square of side $2r$: $A_{\text{square}} = (2r)^2 = 4r^2$.
- Favourable area of the inscribed circle of radius r: $A_{\text{circle}} = \pi r^2$.
- Probability:

$$P(\text{lands inside circle}) = \frac{A_{\text{circle}}}{A_{\text{square}}} = \frac{\pi r^2}{4r^2} = \frac{\pi}{4}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow$ $\frac{\pi}{4}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): When two dice are thrown together, the probability of getting a sum of 8 is $\frac{5}{36}$.

Reason (R): The pairs giving a sum of 8 are $(2,6), (3,5), (4,4), (5,3), (6,2)$, which are 5 favourable outcomes out of 36.

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**

(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**

(c) **(A) is true, but (R) is false.**

(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Reason (R):** Two dice have $n(S) = 36$. Pairs summing to 8 are: $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$ $\implies$ 5 outcomes. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** $P(\text{sum is } 8) = \frac{n(E)}{n(S)} = \frac{5}{36}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** Reason (R) lists the exact favourable outcomes and total sample space that prove Assertion (A). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A number is chosen at random from the numbers $-3, -2, -1, 0, 1, 2, 3$. What is the probability that $x^2 < 4$?

Sample Space: $S = \{-3, -2, -1, 0, 1, 2, 3\} \implies n(S) = 7$ numbers.

Evaluation of $x^2 < 4$:

- For $x = -3$: $(-3)^2 = 9 \geq 4$ (not favourable)
- For $x = -2$: $(-2)^2 = 4 \geq 4$ (strict inequality fails)
- For $x = -1, 0, 1$: $(-1)^2 = 1 < 4, \quad 0^2 = 0 < 4, \quad 1^2 = 1 < 4$ (favourable)
- For $x = 2, 3$: $2^2 = 4 \geq 4, \quad 3^2 = 9 \geq 4$ (not favourable)

Favourable values: $E = \{-1, 0, 1\} \implies n(E) = 3$.

$$P(x^2 < 4) = \frac{n(E)}{n(S)} = \frac{3}{7}$$

Answer: $\mathbf{P(x^2 < 4) = \frac{3}{7}}$

Question 7 [2 Marks]

A letter of the English alphabet is chosen at random. Determine the probability that the chosen letter is: (i) a vowel (ii) a consonant.

Sample Space: Total letters in English alphabet: $n(S) = 26$.

(i) A vowel: Vowels are {A, E, I, O, U} $\implies n(\text{vowels}) = 5 \implies P = \frac{5}{26}$.

Answer (i): $\mathbf{P(\text{vowel}) = \frac{5}{26}}$

(ii) A consonant: Consonants $= 26 - 5 = 21 \implies P = \frac{21}{26}$.

Answer (ii): $\mathbf{P(\text{consonant}) = \frac{21}{26}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

A single die is thrown. Find the probability of getting: (i) a multiple of 2 or 3 (ii) a multiple of 2 and 3 (iii) a number neither divisible by 2 nor by 3.

Sample Space: $S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$.

(i) Multiple of 2 or 3: Multiples of 2 are $\{2, 4, 6\}$; multiples of 3 are $\{3, 6\}$. Union: $\{2, 3, 4, 6\} \implies n(E_1) = 4$:

$$P(\text{multiple of 2 or 3}) = \frac{4}{6} = \frac{2}{3}$$

Answer (i): $\mathbf{P(\text{multiple of 2 or 3}) = \frac{2}{3}}$

(ii) Multiple of 2 and 3: Must be divisible by $\text{LCM}(2, 3) = 6 \implies \{6\} \implies n(E_2) = 1$:

$$P(\text{multiple of 2 and 3}) = \frac{1}{6}$$

Answer (ii): $\mathbf{P(\text{multiple of 2 and 3}) = \frac{1}{6}}$

(iii) Neither divisible by 2 nor by 3: Numbers not divisible by 2 or 3 are $\{1, 5\} \implies n(E_3) = 2$:

$$P(\text{neither 2 nor 3}) = \frac{2}{6} = \frac{1}{3} \quad \left[\text{or } 1 - \frac{2}{3} = \frac{1}{3}\right]$$

Answer (iii): $\mathbf{P(\text{neither divisible by 2 nor 3}) = \frac{1}{3}}$

Question 9 [3 Marks]

Harpreet tosses two different coins simultaneously (one of Rs. 1 and the other of Rs. 2). What is the probability that she gets: (i) at least one head? (ii) at most one tail? (iii) no head?

Sample Space: $S = \{\text{HH, HT, TH, TT}\} \implies n(S) = 4$ equally likely outcomes.

(i) At least one head: Outcomes with 1 or 2 heads: $\{\text{HT, TH, HH}\} \implies n(E_1) = 3 \implies P = \frac{3}{4} = 0.75$.

Answer (i): $\mathbf{P(\text{at least one head}) = \frac{3}{4} = 0.75}$

(ii) At most one tail: Outcomes with 0 or 1 tail: $\{\text{HH, HT, TH}\} \implies n(E_2) = 3 \implies P = \frac{3}{4} = 0.75$.

Answer (ii): $\mathbf{P(\text{at most one tail}) = \frac{3}{4} = 0.75}$

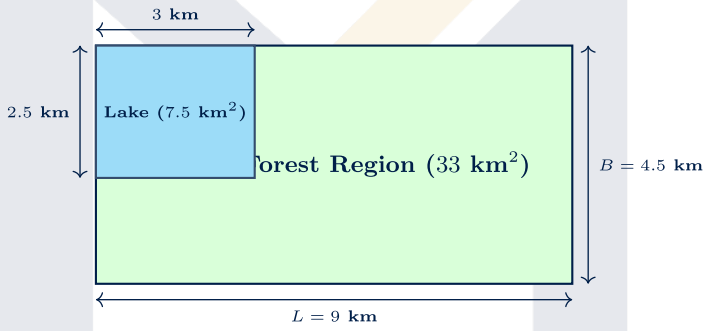
(iii) No head: Both coins show tails: $\{\text{TT}\} \implies n(E_3) = 1 \implies P = \frac{1}{4} = 0.25$.

Answer (iii): $\mathbf{P(\text{no head}) = \frac{1}{4} = 0.25}$

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [10 Marks]

Question 10 [5 Marks]

A missing helicopter is reported to have crashed somewhere in the rectangular region of length 9 km and breadth 4.5 km. Inside this rectangular region, there is a lake of length 3 km and breadth 2.5 km located at one corner, and the rest is forest.



(i) Total area of rectangular region: $A_{\text{total}} = 9 \times 4.5 = 40.5 \text{ km}^2$.

Answer (i): $\mathbf{A_{total} = 40.5 \text{ km}^2}$

(ii) Area of the lake: $A_{\text{lake}} = 3 \times 2.5 = 7.5 \text{ km}^2$.

Answer (ii): $\mathbf{A_{lake} = 7.5 \text{ km}^2}$

(iii) Probability helicopter crashed inside lake: $P = \frac{A_{\text{lake}}}{A_{\text{total}}} = \frac{7.5}{40.5} = \frac{75}{405} = \frac{5}{27}$.

Answer (iii): $\mathbf{P(\text{crashed in lake}) = \frac{5}{27}}$

(iv) Probability helicopter crashed in forest: $A_{\text{forest}} = 40.5 - 7.5 = 33 \text{ km}^2 \implies P = \frac{33}{40.5} = \frac{22}{27}$.

Answer (iv): $\mathbf{P(\text{crashed in forest}) = \frac{22}{27}}$

(v) Fraction of area covered in 3 days: Rate $= 4.5 \text{ km}^2/\text{day} \implies 3 \times 4.5 = 13.5 \text{ km}^2 \implies$ Fraction $= \frac{13.5}{40.5} = \frac{1}{3}$.

Answer (v): $\mathbf{\text{Fraction Covered} = \frac{1}{3}}$

Question 11 (Two Dice HOTS Analysis) [5 Marks]

In a game of two standard dice, calculate the probability of the following outcomes:
(i) both dice show odd numbers (ii) at least one die shows a 5 (iii) neither die shows a 6 (iv) the sum of the numbers is divisible by 4 (v) the product of the numbers is greater than 16.

Sample Space: When two standard dice are rolled, total elementary outcomes: $n(S) = 6 \times 6 = 36$.

(i) Both dice show odd numbers [1 Mark]

Odd numbers on a die are $\{1, 3, 5\}$ (3 choices per die) $\implies n(E_1) = 3 \times 3 = 9$:

$$E_1 = \{(1,1),(1,3),(1,5),(3,1),(3,3),(3,5),(5,1),(5,3),(5,5)\} \implies P(\text{both odd}) = \frac{9}{36} = \frac{1}{4} = 0.25$$

Answer (i): P(both dice show odd numbers) $= \frac{1}{4} = 0.25$

(ii) At least one die shows a 5 [1 Mark]

Outcomes with 5 on the first die: $\{(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)\}$ (6 pairs).
Outcomes with 5 on the second die: $\{(1,5),(2,5),(3,5),(4,5),(6,5)\}$ (5 pairs) $\implies n(E_2) = 6 + 5 = 11$:

$$P(\text{at least one die shows 5}) = \frac{11}{36}$$

Answer (ii): P(at least one die shows 5) $= \frac{11}{36}$

(iii) Neither die shows a 6 [1 Mark]

Each die can show any number from $\{1, 2, 3, 4, 5\}$ (5 choices each) $\implies n(E_3) = 5 \times 5 = 25$:

$$P(\text{neither die shows 6}) = \frac{25}{36} \quad \left[\text{or via complement: } 1 - \frac{11}{36} = \frac{25}{36}\right]$$

Answer (iii): P(neither die shows 6) $= \frac{25}{36}$

(iv) The sum of the numbers is divisible by 4 [1 Mark]

Possible sums divisible by 4 are 4, 8, and 12:

- Sum $= 4$: $\{(1,3),(2,2),(3,1)\} \implies 3$ pairs
- Sum $= 8$: $\{(2,6),(3,5),(4,4),(5,3),(6,2)\} \implies 5$ pairs
- Sum $= 12$: $\{(6,6)\} \implies 1$ pair

Total favourable outcomes $n(E_4) = 3 + 5 + 1 = 9 \implies P = \frac{9}{36} = \frac{1}{4} = 0.25$.

Answer (iv): P(sum divisible by 4) $= \frac{9}{36} = \frac{1}{4} = 0.25$

(v) The product of the numbers is greater than 16 [1 Mark]

Pairs (x, y) satisfying $x \cdot y > 16$:

- $x = 3$: $(3,6) \implies 1$ pair
- $x = 4$: $(4,5),(4,6) \implies 2$ pairs
- $x = 5$: $(5,4),(5,5),(5,6) \implies 3$ pairs
- $x = 6$: $(6,3),(6,4),(6,5),(6,6) \implies 4$ pairs

Total favourable outcomes $n(E_5) = 1 + 2 + 3 + 4 = 10 \implies P = \frac{10}{36} = \frac{5}{18}$.

Answer (v): P(product > 16) $= \frac{10}{36} = \frac{5}{18}$

— END OF WORKED SOLUTIONS (TEST–11) · MATHS MANTHAN TUITION CLASSES —